CAD Model of Trachea, Head, and Neck for DoD Endotracheal Intubation on the Battlefield

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Overview

The objective of this project is to design a CAD model of the upper respiratory system - including the trachea, nasal cavity, oral cavity, larynx, etc. - which can be used to test a device. The device is a self-deploying, instrumented, shape-memory polymer endotracheal tube for non-expert, autonomous emergency intubation and monitoring of thoracic injuries in battlefield conditions. The model must match the dimensions of a human in order to provide accurate data and test results. The model must also be 3D printed for tests, and therefore must be designed structurally.

Research

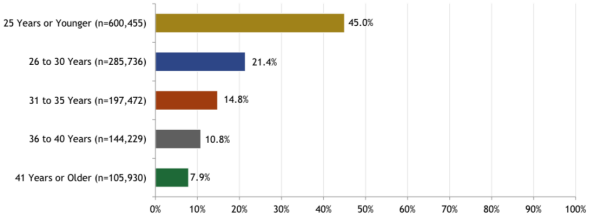
Demographic Selection

Using the 2020 Demographics provided by the Department of Defense (DoD) we analyzed the different demographic groups that were found in the military while also researching the groups that were most commonly needed for endotracheal intubation. The conclusion was that an ideal person to design the tracheal model for was a 25-year-old white male. The data to the right indicates that nearly 50% of the active duty members in the military are below the age of 25, and half are above the age of 25. An argument can be made that endotracheal intubation is needed more in elderly folks, older than 60; however, considering that fewer than 10% of the army is 41 years older, it is illogical to design it for the elderly population. When it comes to choosing the gender, the choice was simple, more than 80% of the military comprises men, making it logical to first test the device on a man. Similar to the selection of the gender, there is one clear population that has more people enlisted in the army, and that is white people. White people make up nearly 70% of the military, with the next closest being Black or African American with nearly 20%. Additionally, both white and male people are more likely to need endotracheal intubation than any other gender or race.

Age

2.41 | Age of Active Duty Members (N=1,333,822)

Nearly one-half (45.0%) of Active Duty members are 25 years of age or younger.

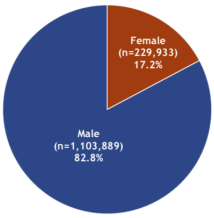


Note: 220 Active Duty members with unknown age are categorized as 25 years or younger.
Note: Displayed percentages may not total 100% due to rounding.
Source: DMDC Active Duty Military Personnel Master File (September 2020)

Gender

2.09 | Gender of Active Duty Members (N=1,333,822)

There are 229,933 (17.2%) female Active Duty members and 1,103,889 (82.8%) male Active Duty members.

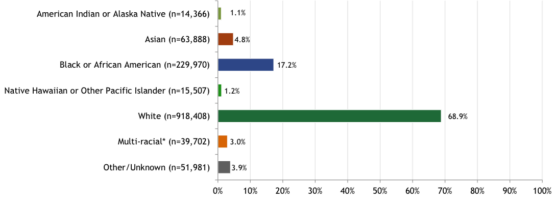


Note: Displayed percentages may not total 100% due to rounding.
Source: DMDC Active Duty Military Personnel Master File (September 2020)

Race/Ethnicity

2.21 | Race of Active Duty Members (N=1,333,822)

Members who report themselves as White make up the highest percentage of Active Duty members (68.9%), while Black or African American members represent 17.2 percent. Asian, Native Hawaiian or Other Pacific Islander, American Indian or Alaska Native, and Multi-racial members each make up less than 5.0 percent of Active Duty members.



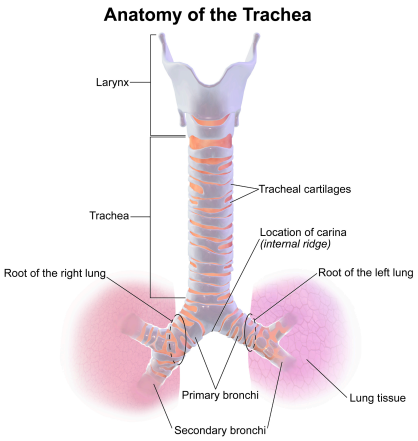
* The Army does not report "Multi-racial."
Note: Displayed percentages may not total 100% due to rounding.
Source: DMDC Active Duty Military Personnel Master File (September 2020)

Anatomy

Trachea Anatomy

The trachea, also known as the windpipe, is a long cartilagionous tube part of the lower respiratory tract. It is in the middle part of the lower respiratory tract (the larynx and downwards), which is made of three parts: the larynx, trachea, and bronchi. The rower respiratory system begins at the 6th cervical vertebra and ends at the 5th - 7th thoracic vertebrae.

As the trachea descends, it splits into two parts which are “mini trachea”, called the bronchi. It is about 9-15 cm long and 2-2.5 cm in diameter. The trachea is a consititutively patent structure with circumferential cartilaginous rings. The annular ligaments and trachealis muscle connect the horseshoe-shaped rings, maintaining airway patency and allowing for flexibility during inspiration and expiration. Located directly in front of the esophagus, the tracheal wall features a posterior membranous part. It consists of four distinct layers: the innermost Tunica Mucosa, the middle Tela Submucosa (containing loose connective tissue and blood vessels), and the outermost Tunica Adventitia (dense connective tissue). At its base, the trachea splits into a tracheal bifurcation, marked internally by the carina, a ridge that separates the bronchi.



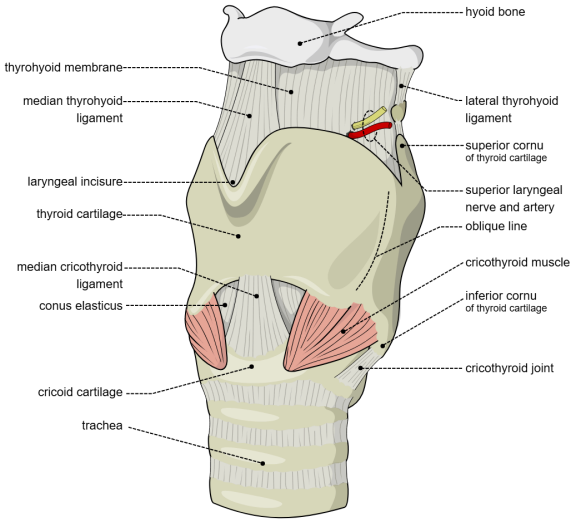
For an accurate CAD model, we needed to supplement the basic dimensions with some precise geometric data and structural counts. Specifically, we need to incorporate the standard quantity of 16 to 20 C- shaped cartilginous rings, defining each ring with an approximate height of 4 mm and a spacing of 2 to 3 mm via the annular ligaments. We must also account for the asymmetrical cross-section where the lateral diameter is typically larger (around 2.5 cm) than the anterior-posterior depth (around 2.0 cm), with the flat posterior membranous wall. Furthermore, we have to include precise geometry of the bifurcation at the carina, where the trachea splits into the main bronchi at an internal angle of roughly 50 to 70 degrees, ensuring the right bronchus is modeled as wider and more vertical than the left. Finally, to sweep or loft the final 3D shape correctly, you need to map a central 3D coordinate curve or centerline path that captures the trachea's natural, gentle anterior concavity as it follows the curvature of the spine into the thoracic cavity.

Larynx Anatomy

The larynx, commonly called the voice box, a hollow, tubular structure in the anterior neck that cats as a switch between the respiratory and digestive tracts. Its primary functions include protecting the airway from aspirated food, regulating airflow for breathing, and producing the sound needed for speech. The larynx acts as the critical midpoint and gating mechanism of the respiratory tract, physically linking the upper and lower respiratory systems. The larynx transitions air between the following sequential respiratory structures: the pharynx, where food and air travel before separating; and trachea, where the lower border cricoid cartilage marks the entrance to the lower respiratory tract.

In terms of the larynx, it is located relatively to many different structures in neck. It sits directly behind infrahyoid muscles and skin of the anterior neck. It also sits directly in front of the esophagus, and

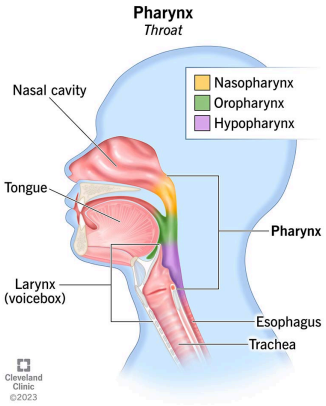
the rigid cartilages protect the airway from collapsing when food passes down. Furthermore to the side they are carotid sheaths, which contain major blodd vessels, and thyroid gland. It spans vertically from the C3 to C6 vertebral levels.



It is structurally supported by nine cartilages – three unpaired (thyroid, cricoid, epiglottis) and three paired (arytenoids, corniculates, cuneiforms) – with ligaments like the thyrohyoid and cricothyroid. On the inside, the larynx is divided into three segments: the supraglottis, glottis, and subglottis. The supraglottis sits above the true vocal cords, which contains the epiglottis and false vocal folds, while the glottis has the true vocal cords that drive sound production during expiration. The subglottis extends from the glottis to the cricoid cartilage. For movement and sensation, the larynx relies on a system of extrinsic and intrinsic muscles and is primarily supplied by the superior and inferior laryngeal arteries. The superior laryngeal artery, branching from the superior thyroid artery off the external carotid, is the main contributor to the upper laryngeal structures.

Pharynx Anatomy

The pharynx, which is a muscular and cone-shaped tube, that marks the intersection of the respiratory and digestive system. It starts from the base of the skull and goes down to the level of the sixth cervical vertebra, which is where it divides into the esophogus and larynx. The pharynx is divided into three anatomical zones: the nasopharynx, the oropharynx, and the laryngopharynx. The top one, the nasopharynx, is the passage for air, while to lower two sections are for the air and solid foods and liquids which are swallowed.



Because of the fact that it handles both food and air, the pharynx has many critical immune defense tissues and pressure regulating structures. In the nasopharynx, there are the adenoids and the openings to the Eustachian tubes, which connect to the middle ears, in order to balance air pressure. As you move down to the oropharynx, there are the prominent palatine tonsils, which guard against pathogens. The oropharynx and laryngopharynx has a multi-layered tissue layer. The specialized lining protects the throat from the friction and wear caused by passing food.

Oral Cavity

The oral cavity, commonly kwnon as the mouth, is the secondary entrance to the respiratory system. It is surrounded by the lips, cheeks, tongue, and the plate, which is divided into the hard at the front and the soft palate at the back. Under normal conditions, most people breathe through their nose. However the month is another pathway for air especially when the nose is blocked because illness,

allergies, or other conditions. Furthermore, during exercise or other activities that require more air and oxygen, the mouth becomes an important airway. Since it is much wider than the masal passages, it allows for a greater amount of air to move in and out of the lungs with less resistance. However, breathing through the mouth also has a disadvantage because the air is not filtered, warmed or moistened as well as it is when passing through the nose.

Nose & Nasal Cavity

The nose and nasal cavity are the main entrance to the respiratory system and are responsible for cleaning and preparing the air before it will reach the lungs. The air enters through the nostrils and passes into the nasal cavity, which is divided into right and left sides by a wall called the nasal septum. Inside the nostrils is the nasal vestibule, which contains coarse hairs called vibrissae. These heair trap larger particles such as dust, dirt, and insects before they travel deeper into the respiratory system.

As the air moves through the nasal cavity, it passes over three pairs of curved bones called the nasal conchae, or turbinates. These bones create turbulence, which causes the air to come into contect with the warm, moist lining of the nasal cavity. This lining contains mucus that also traps harmful particles such as dust and bacteria. Tiny hair-like structures called cilia then move the mucus toward the back of the throat, where it is swallowed and destroyed by stomach acid.

CAD Software Selection

Once a solid research base is gained, the next stage was selecting a CAD model that could be easily used for this project. The three CAD softwares that were considered are Solidworks, OnShape, Blender This was selected based on the following criteria:

- ☐ Cost (*Is this software free or does it require a payment?*)
- ☐ Organic Modeling (*Does this software have features for organic modeling?*)
- ☐ Simplicity (*How easy is this software to use and learn?*)
- ☐ Experience (*How much experience do I have with this software?*)
- ☐ Avaiability (*Can this be accessed on a regular computer or does it need a certain GPU or RAM?*)
- ☐ Features (*Does it have extensive features or is the software limited?)*

Solidworks

Solidworks is a 3D parametric solid modeling computer-aided design (CAD) and computer aided engineering (CAE) software. It is used by many designers, engineers, and manufacturers to build, simulate, and document digital prototypes before being produced in the real world. Solidworks using parametric modeling, meaning instead of just drawing lines, you define mathematical rules, dimensions, and relationships. If you change a single dimension, the rest of the model automatically updates to match.

It has three design environments for the three key stages of design which are as follows: parts, where you design individual 3D components; assemblies, where to put those individual parts together to see how they fit and interact with one another; and drawings, which automatically generate 2D technical

blueprints for manufacturing and production. It also includes tools to test your design for stress, strain, weight, and motion without needing a physical prototype.

Category	**Rating**
Cost	☑☑☑☑☑☑☑☑☑☐
Organic Modeling	☑☑☑☐☐☐☐☐☐☐
Simplicity	☑☑☑☑☑☑☑☐☐☐
Experience	☑☑☑☑☑☑☐☐☐☐
Avaiability	☑☑☑☑☑☐☐☐☐☐
Features	☑☑☑☑☑☑☑☑☑☐

OnShape

Onshape is a cloud-native 3D parametric computer-aided (CAD) and produce data management (PDM) platform. It is used by engineers and designers to build products, collaborate in real time, and manage data entirely through a web browser or mobile app.

Onshape has no software to install or update. The entire platform runs entirely in a web browser or via mobile apps, utilizing cloud servers to hand heavy processing. It eliminates the need for separate data management software. It works in a similar manner to Google Docs, automatically saving every change and allowing teams to view a complete edit history and change designs without overwriting the files. It has real-time collaboration where multiple designers can work on the exact same part or assembly simultaneously. You can see teammates' cursors, edits, and comments live as they happen. Furthermore, unlike traditional CAD where each file holds only one parts, Onshape allows the design of multiple interrelated parts together in a single workspace. This makes it much easier to ensure parts fit together correctly from the start.

Category	**Rating**
Cost	☑☑☑☑☑☑☑☐☐☐
Organic Modeling	☑☑☐☐☐☐☐☐☐☐
Simplicity	☑☑☑☑☑☑☑☑☑☐
Experience	☑☑☑☐☐☐☐☐☐☐
Avaiability	☑☑☑☑☑☑☑☑☑☑

Features	☑☑☑☑☑☑☐☐☐☐

Blender



Blender functions as a subdivision surface (SubD) and digital sculpting CAD tool for organic modeling when used for physical production. While it lacks certain history-based parametric constraints, it is used by designers to create complex, fluid shapes which are difficult to build in a traditional engineering software.

It acts like digital clay, allowing you to pull, push, and smooth meshes using brush strokes to create organic textures and biological forms. You can build simple geometric cages that the software can automatically smooth out into continuous, flowing surfaces, which is ideal for ergonomic handles, footwear, and consumer products. Furthermore, by enabling built-in or community tools such as CAD Sketcher or Measureit, you can add precise dimensional constraints, snapping, and measurements to organic shapes. It also includes a 3D print toolbox to check meshes for watertightness, wall thickness, and manifold errors before exporting STL or OBJ files for manufacturing.

Category	**Rating**
Cost	☑☑☑☑☑☑☑☑☐☐
Organic Modeling	☑☑☑☑☑☑☑☑☑☐
Simplicity	☑☑☑☑☑☐☐☐☐☐
Experience	☑☑☐☐☐☐☐☐☐☐
Avaiability	☑☑☑☑☑☐☐☐☐☐
Features	☑☑☑☑☑☑☑☑☑☑

Final Selection: Blender

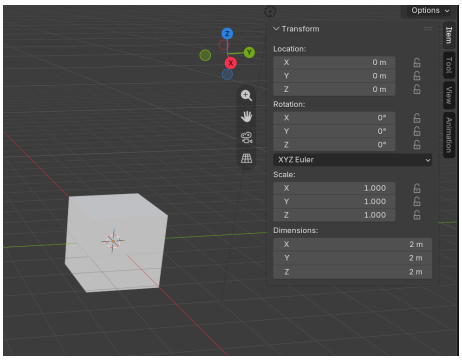
We concluded that Blender would be the best CAD software for the project that we were doing because of its low cost and features for organic shapes. While we had more experience and certifications with SolidWorks and Onshape, Blender provides more benefits and support in the long run. Onshape and Solidworks are primarily used for geometrical shapes and mechanical parts, rather than the biological and anatomical shapes that we strive to make.

Software Learning

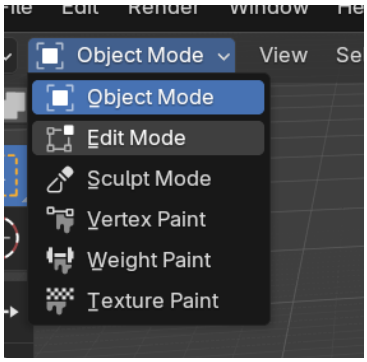
As an absolute beginner to Blender, in order to model biological structures requires a process to learn the basics to specialized modifiers and curves. The most effective path for this software learning involves learning to manipulate shapes, extrude tubes, and sculpt soft anatomy.

Stage 1: The Foundations

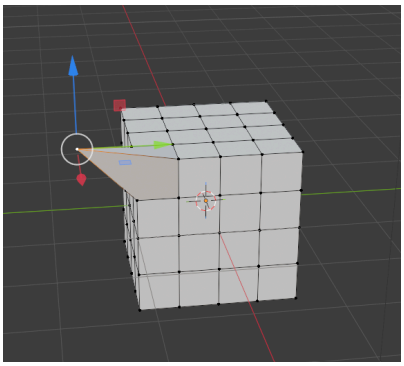
Before we can start building realistic models of the tracheal or vocal cords, we have to learn how to actually move around in Blender's 3D world without getting lost or feeling overwhelmed. This stage is focused on rearning how to rotate our view, move across the screen, and zoom in on objects so that we won't lose track of our work. We will also learn how to generate basic shapes like cubes and cylinders, and practice the three major controls: moving them around, rotating them, and scaling them to make them bigger or smaller. The most important skill that we focused on learning in this stage is switching between the Tab key and to switch into Edit Mode. This allows us to click on the individual points, lines, and faces of a shape so we can stretch, move, and mold it. In this stage, we used YouTube tutorials, which taught us the basic steps in an interactive way.



Use the transform and scale tool.



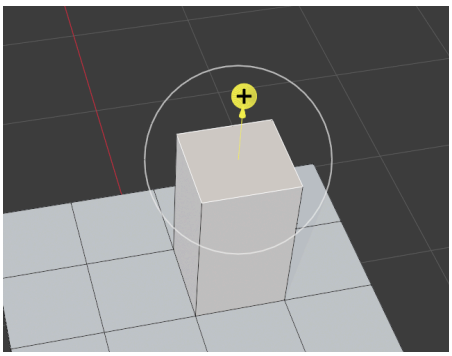
Toggle between object and edit mode.



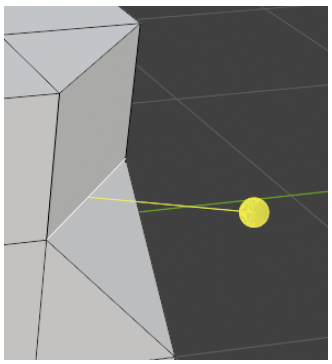
Move points, edges, and faces

Stage 2: Hard-Surface Modeling

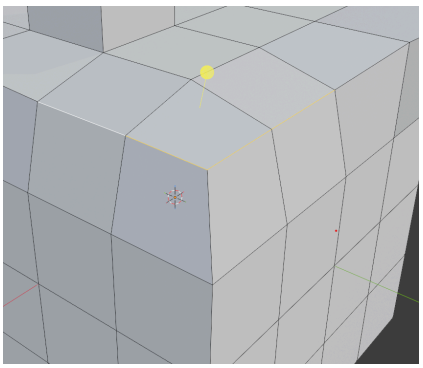
Once we were good with navigating the software, we can start learning how to build the basic frame work of the upper respiratory system using simple geometric shapes. The airway is in simple terms a network of hollow, branching tubes, so we spent a lot of time learning the Extrude tool, which lets us grab a flat face of a cylinder and create a long pipe. We also learned how to use more advanced features that will allow us to bend and curve flat, hard-surface pieces. Another thing we spent a lot of time learning how to do is moving points, sides, and faces to make them more accurate to anatomical shapes. This would become essential as we continued to mold and fix our model for it to be 3D printed.



Extrude faces to create 3D shapes



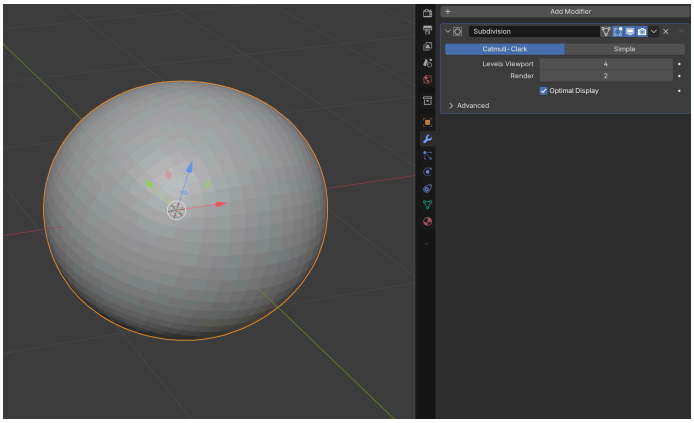
Bevel edges for more intricate shapes



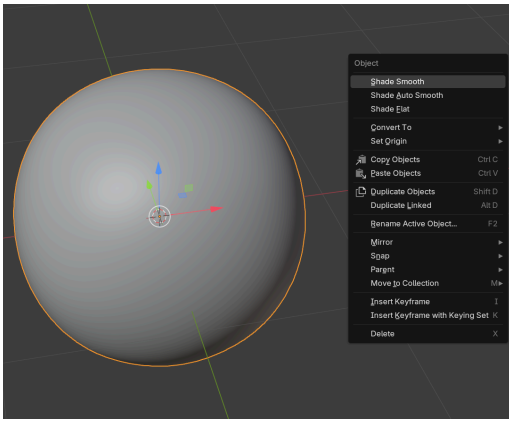
Smooth edges and faces for a more clean look

Stage 3: Organic Modeling

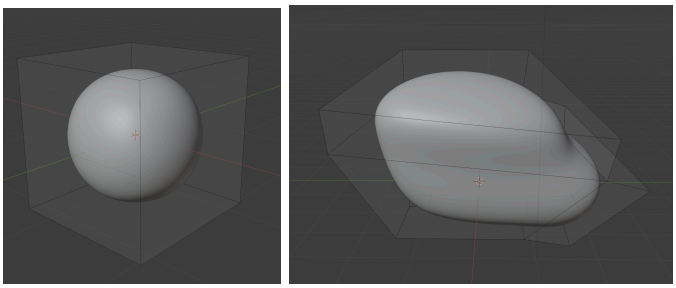
After were comfortable with hard-surface modeling, we begin focusing on organic modeling, which we will use to create the smooth natural shapes used for anatomical structures. Because the upper respiratory system will have curved structures such as the larynx, pharynx, nasal cavity, and oral cavity, we had to learn techniques which allowed us to create natural and flowing shapes rather than sharp, rigid shapes. We learned how to use loop cuts and the subdividing surface modifier, giving us more control over the detail and shape of each structure. In summary, this stage helped us transform simple cylinders and cubes into a realistic anatomical structure, which will be helpful when completing the respiratory system.



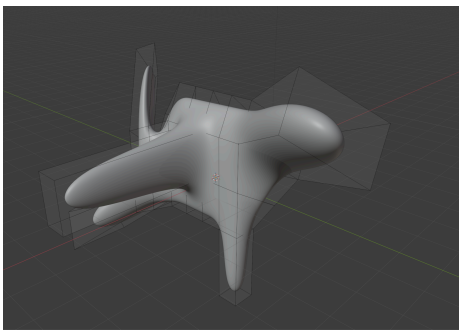
Apply the Subdivision Surface Modifier to smooth geometry.



Use Shade Smooth for a smooth finish



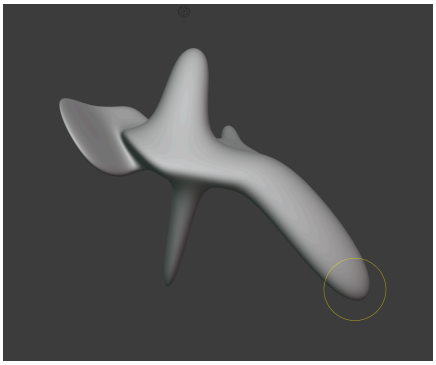
Shape complex anatomy by adjusting individual vertices.



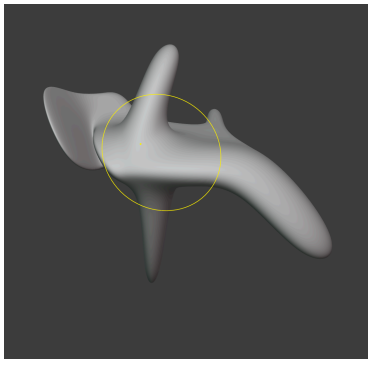
Add Loop Cuts to increase modeling control.

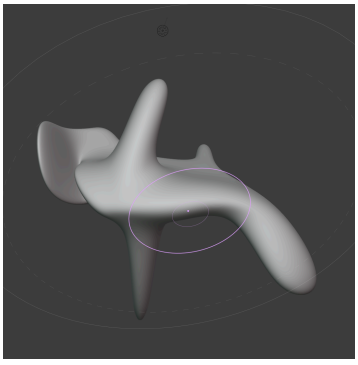
Stage 4: Digital Sculpting

After we understood how to make basic shapes of each part, we started learning more advanced features in digital sculpting to make it look more realistic and add more unique shapes. Sculpting makes the part you are working on like clay, rather than moving one point, edge, or face. We learned how to use the grab brush to move large areas, the smooth brush to get rid of rough spots, and the inflate brush to make certain parts stick out more. This allowed us to make the upper respiratory system look much more realistic and anatomically accurate. We also learned that sculpting works best after the basic shape is already finished, while Edit Mode is still better for making exact changes. By the end of this stage, we were able to clean up our model, smooth out the connections between different parts, and make it look much more realistic before getting it ready for 3D printing.



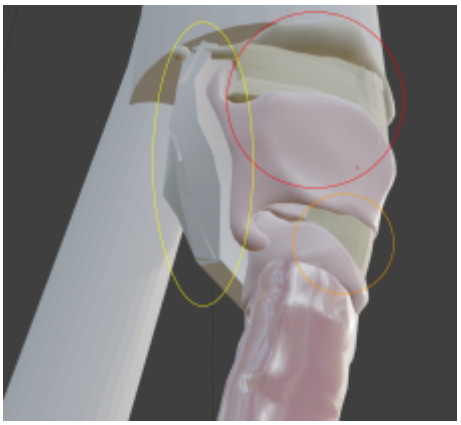
Navigate and use Sculpt Mode.

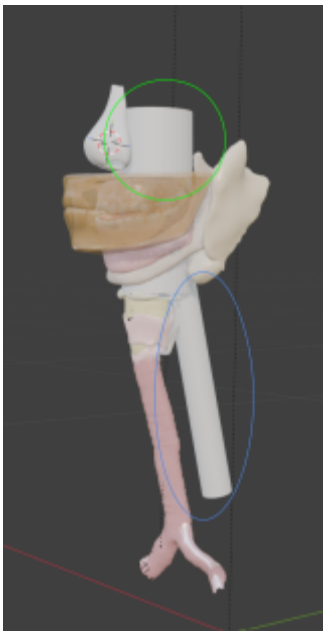




Use the Grab, Smooth, Inflate, and Crease brushes.

Initial Designing





The initial design was focused on importing already designed parts from sites such as the NIH database. This included parts of the larynx, trachea, mouth (teeth, skin, tongue, and hard/soft palate), nose, and jaw.

From there, we started by designing the thyrohyoid membrane and the median thyrohyoid ligament (Circled in Red) using only the grab and rotate features. After that, we designed the median cricothyroid ligament and the conus elasticus (Circled in Orange), which were very similar to the thyrohyoid membrane. Completing these four components meant that the posterior part of the pharynx was done. For the anterior component (Circled in Yellow), we simply created a back panel for the trachea, focusing on maintaining an anatomically accurate shape.

To complete the initial design phase, we added the pharynx (Circled in Green) and esophagus (Circled in Blue) as two large cylinders that passed straight through the rest of our model. As will be explained earlier, this caused many issues when attempting to design the nasal cavity and mesh the model together.

Design Improvement

Design Refinement

Key Results / Final Specifications

Next Steps

Works Cited

https://pmc.ncbi.nlm.nih.gov/articles/PMC5771073/
https://www.annalscts.com/article/view/16463/html
https://www.youtube.com/watch?v=GfuGw_uCdig&t=18s
https://www.youtube.com/watch?v=J3WS7yNfK5U
https://www.youtube.com/watch?v=5F8XdE1d0xk